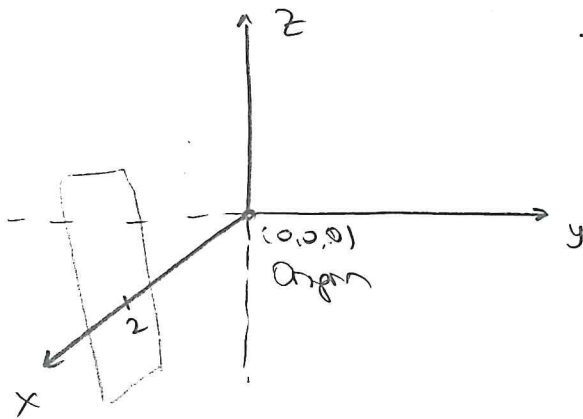


CALCULUS (II)

WEEK IV

VECTORS AND COORDINATE GEOMETRY IN 3-SPACE



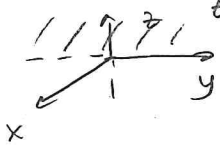
The three coordinate planes $x=0$, $y=0$ and $z=0$ divide space into eight cells called octants. The octant in which the point coordinates are all positive is called First octant ($x>0, y>0, z>0$); there is no convention for numbering the other seven octants.

→ To write equations for planes perpendicular to coordinate axes, we name the coordinate's values.

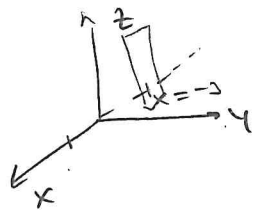
→ The plane $x=2$ is the plane perpendicular to the x -axis at $x=2$. The plane $y=3$ is the plane perpendicular to the y -axis at $y=3$.

Ex 7 Interpret the following equations and inequalities geometrically.

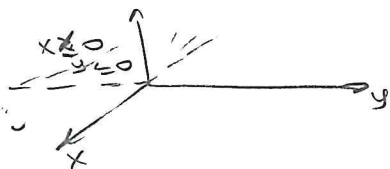
(a) $z \geq 0$. The half-space consisting of the points on and above the xy -plane.



(b) $x = -3$. The plane perpendicular to the x -axis at $x = -3$. This plane lies parallel to the yz -plane and 3 units behind it.



(c) $z = 0, x \leq 0, y \geq 0$. The second quadrant of the xy plane.



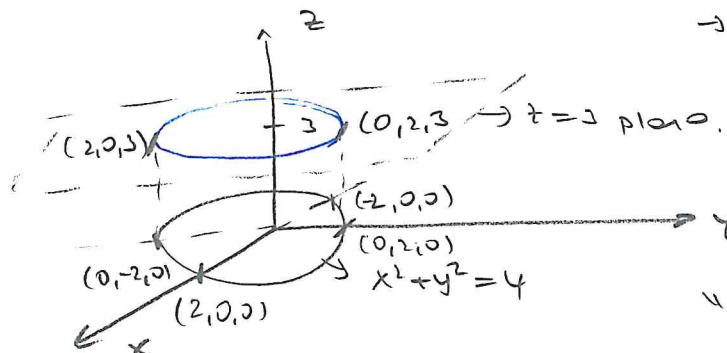
(d) $x \geq 0, y \geq 0, z \geq 0$. The first octant.

② $-1 \leq y \leq 1$ The slab between the planes $y = -1$ and $y = 1$ (planes included)

④ $y = -2, z = 2$ The line in which the planes $y = -2$ and $z = 2$ intersect. Alternatively, the line through the point $(0, -2, 2)$ parallel to the x -axis.

Ex ⑦ What points $P(x, y, z)$ satisfy the equations

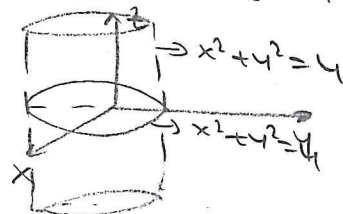
$$x^2 + y^2 = 4 \quad \text{and} \quad z = 3$$



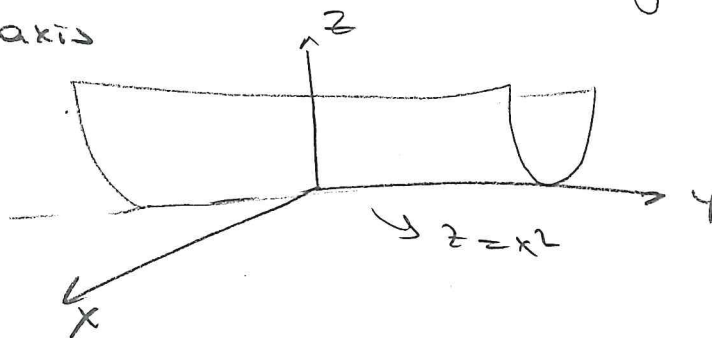
→ The points lie in the horizontal plane $z = 3$ and in this plane, make up the circle $x^2 + y^2 = 4$. We call this set of points

"the circle $x^2 + y^2 = 4$ in the plane $z = 3$ " or simply "the circle $x^2 + y^2 = 4, z = 3$ ".

⑤ The equation $x^2 + y^2 = 4$ represents all points on the vertical circular cylinder containing the circle with equation $x^2 + y^2 = 4$ in the xy -plane. This cylinder has radius 2 and axis along the z -axis.



③ The equation $z = x^2$ represents all points which coordinates (x, y, x^2) . This surface is a parabolic cylinder tangent to the xy -plane the y -axis.



The Distance Between $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is

$$|P_1 P_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

The standard Equation for the sphere of Radius a and center (x_0, y_0, z_0)

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2$$

Ex Find the center and radius of the sphere

$$x^2 + y^2 + z^2 + 3x - 4z + 1 = 0$$

$$x^2 + 3x + y^2 + z^2 - 4z + 1 = 0$$

$$\left(x^2 + 3x + \frac{9}{4} - \frac{9}{4}\right) + y^2 + (z^2 - 4z + 4 - 4) + 1 = 0$$

$$\left(x + \frac{3}{2}\right)^2 + y^2 + (z - 2)^2 + 3 - \frac{9}{4} = 0$$

$$\left(x + \frac{3}{2}\right)^2 + y^2 + (z - 2)^2 = \frac{21}{4}$$

center $(-3/2, 0, 2)$ radius $\sqrt{21}/2$

EUCLIDEAN n -SPACE

$$\mathbb{R}^n = \{ (x_1, x_2, \dots, x_n) \mid x_1, x_2, \dots, x_n \in \mathbb{R} \}$$

The distance between points $(x_1, x_2, \dots, x_n)$ and $(y_1, y_2, \dots, y_n)$ is defined to be

$$\sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2 + \dots + (y_n - x_n)^2}$$

Describing Sets in the Plane, 3-space, and n -space

A neighbourhood of a point P in $\mathbb{R}^n$ is a set of the form

$$B_r(P) = \{ Q \in \mathbb{R}^n : \text{distance from } Q \text{ to } P < r \}$$

for some $r > 0$.

For $n=1$, if $p \in \mathbb{R}$, then $B_r(p)$ is the open interval $(p-r, p+r)$ centered at p .

For $n=2$, if $B_r(p)$ is the open disk of radius r centered at p .

For $n=3$, if $B_r(p)$ is the open ball of radius r centered at point p .

Describing Sets in the Plane, 3-Space and n -Space

A set S is open in $\mathbb{R}^n$ if every point of S has a neighbourhood contained in S . Every neighbourhood is itself an open set.

Typically, sets defined by strict inequalities (using $>$ and $<$) are open. Examples in $\mathbb{R}^3$ include the sets of points (x, y, z) satisfying $x+y+z > 2$ or $1 < x < 3$.

→ The whole space $\mathbb{R}^n$ is an open set in itself. For technical reasons, the empty set (containing no points) is also considered to be open (no point in the empty set fails to have a neighbourhood contained in the empty set).

→ The complement S^c , of a set S in $\mathbb{R}^n$ is the set of all points in $\mathbb{R}^n$ do not belong to S . For example, the complement of the set of points (x, y) in $\mathbb{R}^2$ such that $x > 0$ is the set of points for which $x \leq 0$.

→ A set is said to be closed if its complement is open.

Typically the sets defined by nonstrict inequalities (using $\geq$ or $\leq$) are closed.

→ Closed intervals are closed sets in $\mathbb{R}$.

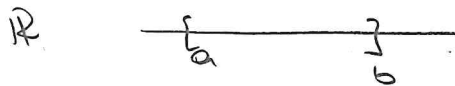
→ Since the whole space and the empty set are both open in $\mathbb{R}^n$ and are complements of each other, they are also both closed. They are the only sets that are both open and closed.

→ A point P is called boundary point of a set S ^{$\nearrow$ (bdry(S))} if every neighbourhood of P contains both points in S and points in S^c .



For $S = [2, 3]$ interval $x=2$ and $x=3$ is boundary points of S because $(2-\epsilon_1, 2+\epsilon_1)$ and $(3-\epsilon_2, 3+\epsilon_2)$ both includes points from S and S^c .

→ A point P is an interior point of a set S if it belongs to S but not to the boundary of S . P is an exterior point of S if it belongs to the complement of S but not to the boundary point of S . The interior, $\text{int}(S)$, and exterior, $\text{ext}(S)$, of S consists of all the interior points and exterior points of S respectively. Both $\text{int}(S)$ and $\text{ext}(S)$ are open sets.

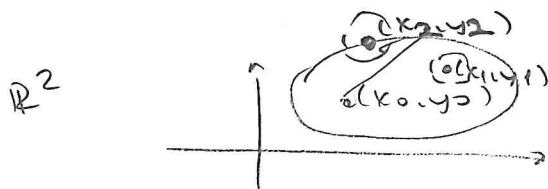


$$S = [a, b]$$

Boundary points $x=a$ $x=b$

$$\text{int}(S) = (a, b)$$

$$\text{ext}(S) = \{x \in \mathbb{R} : x < a, x > b\}$$



$$S: (x-x_0)^2 + (y-y_0)^2 \leq r^2$$

$$\text{int}(S) = \{ (x, y) : (x-x_0)^2 + (y-y_0)^2 < r^2 \}$$

$$\text{bdy}(S) = \text{Boundary}(S) = \{ (x, y) : (x-x_0)^2 + (y-y_0)^2 = r^2 \}$$

$$\text{ext}(S) = \{ (x, y) : (x-x_0)^2 + (y-y_0)^2 > r^2 \}$$

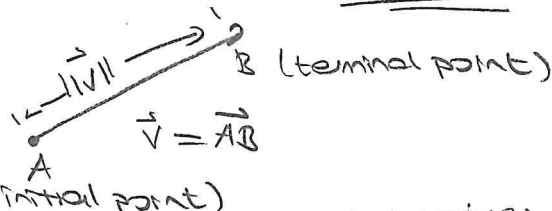
(x_1, y_1) interior point
 (x_2, y_2) boundary point

→ If S is open, then $\text{int}(S) = S$

If S is closed, then $\text{ext}(S) = S^c$

VECTORS

Definition A VECTOR is a quantity that involves both MAGNITUDE (size or length) and DIRECTION. In space $\vec{AB}$ denotes the vector (from the point A to the point B) with initial point A and terminal point B.

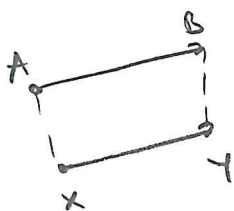


For instance, the velocity of a moving object involves its speed and its direction of motion so is a vector.

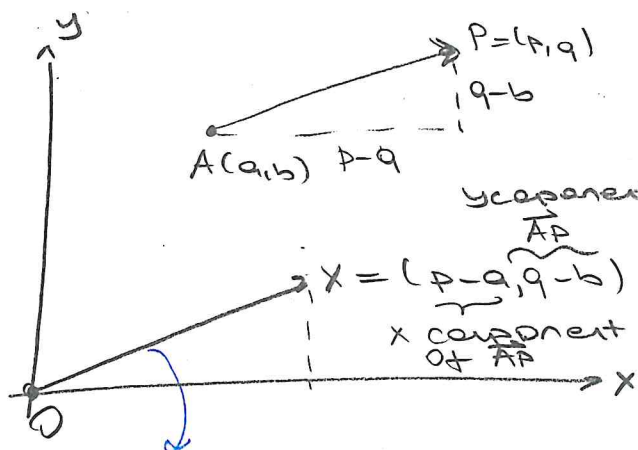
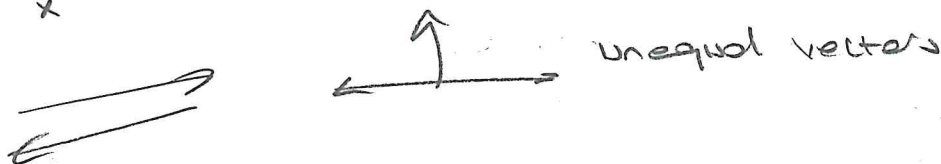
The MAGNITUDE (OR LENGTH) of the vector $\vec{AB}$ is the length of the line segment AB, and denoted by $|\vec{AB}|$ (or $\|\vec{AB}\|$).

Note (x, y) is a point, $\langle x, y \rangle$ is a vector (scalar value + direction)
(Scalar value)

Definition Two vectors are assumed to be EQUAL (or the SAME) if they have the same magnitude and same direction.



$\vec{AB}$ and $\vec{XY}$ are same, equal vectors



$$|\vec{AP}| = \sqrt{(p-a)^2 + (q-b)^2} = |\vec{OX}|$$

$$\text{Slope of } \vec{AP} = \frac{q-b}{p-a} = \text{slope of } \vec{OX}$$

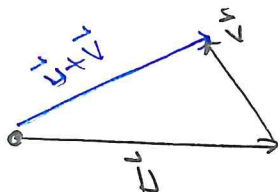
Hence, $\vec{AP} = \vec{OX}$. In general, two vectors are equal if and only if they have the same x components and y components.

$\vec{x}$ is the position vector of the point

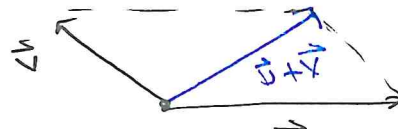
$x = (p-a, q-b)$ (position vector from origin to point)

VECTOR ADDITION

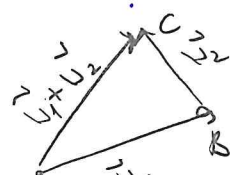
→ Given two vectors $\vec{u}$ and $\vec{v}$, their sum $\vec{u} + \vec{v}$ is defined as follows. If an arrow representing $\vec{v}$ is placed with its tail at the head of an arrow representing $\vec{u}$, then an arrow from the tail of $\vec{u}$ to the head of $\vec{v}$ represents $\vec{u} + \vec{v}$ (Triangle Law)



(Triangle Law)



(Parallelogram Law)



$$\vec{u}_1 + \vec{u}_2 = \vec{AB} + \vec{BC} = \vec{AC}$$

→ Equivalently, if $\vec{u}$ and $\vec{v}$ have tails at the same point, then $\vec{u} + \vec{v}$ is represented by an arrow with its tail at the point and its head at the opposite vertex of the parallelogram spanned by $\vec{u}$ and $\vec{v}$ (Parallelogram Law)

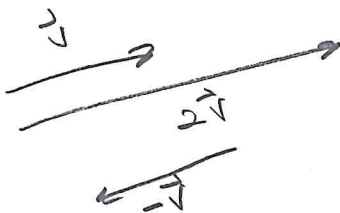
→ Tail Head

SCALAR MULTIPLES OF $\vec{v}$

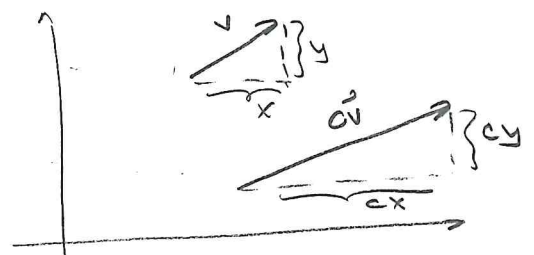
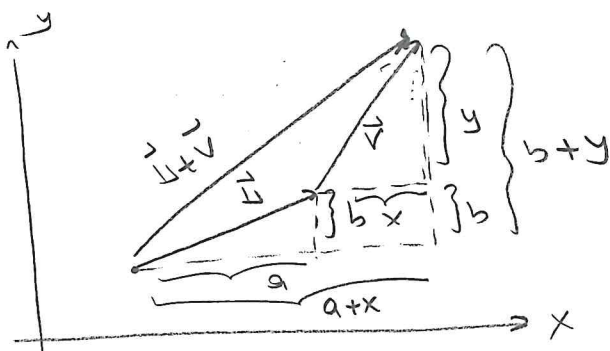
If c is a nonzero real number, the direction of $c\vec{v}$ is same with $\vec{v}$ if c is positive and is opposite

that of $\vec{v}$ if c is negative. The scalar multiple $c\vec{v}$ is

a vector with magnitude $|c|$ times of $\vec{v}$ and direction the same as $\vec{v}$ if $c > 0$, or opposite to that of $\vec{v}$ if $c < 0$.



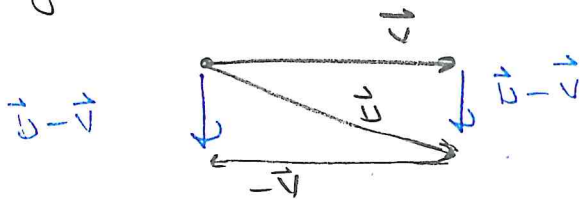
→ If $c = 0$, then $c\vec{v}$ has zero length and therefore no particular direction. It is zero vector, denoted $\vec{0}$



→ Suppose that $\vec{u}$ has components a and b and that $\vec{v}$ has components x and y . Then the components of $\vec{u} + \vec{v}$ are $a+x$ and $b+y$, and $c\vec{v}$ is cx and cy . (7)

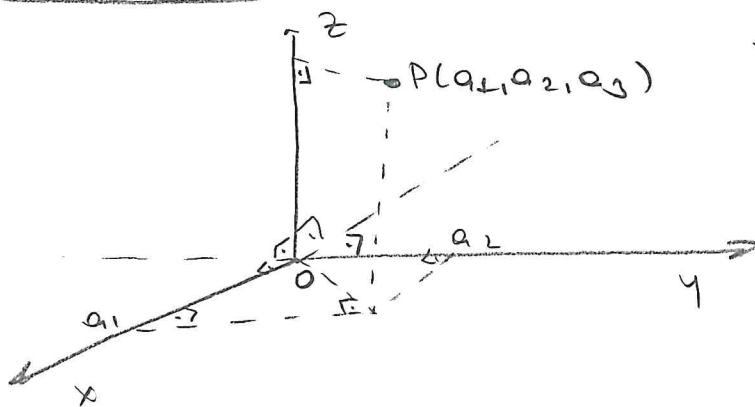
Def The NEGATIVE of a vector $\vec{v}$ is $(-1)\vec{v} = (-\vec{v})$

Def The DIFFERENCE $\vec{u} - \vec{v}$ of $\vec{u}$ and $\vec{v}$ is the vector $\vec{u} + (-\vec{v})$. It is the vector, which when added to $\vec{v}$ gives the vector $\vec{u}$



COMPONENT FORM OF VECTORS

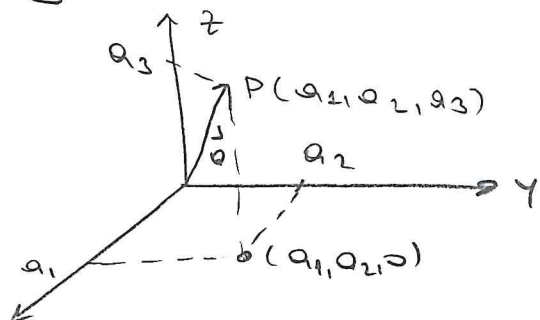
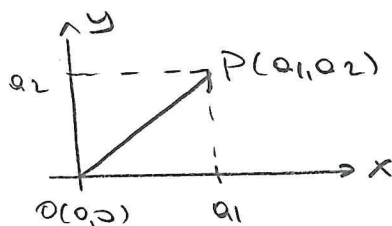
Def In space three perpendicular lines through the same point O , as shown in the figure, form a CARTESIAN or RECTANGULAR coordinate system.



→ The point O is called the origin and the lines are called the COORDINATE AXES.

→ Any point P in space is determined by three real numbers when a rectangular coordinate system is fixed. Those numbers are called the (cartesian) COORDINATES of that points. The coordinates of the origin $O = (0, 0, 0)$

→ Place the initial point of a vector $\vec{a}$ at the origin of a cartesian coordinate system



→ The terminal point P of $\vec{a}$ has coordinates of the form (a_1, a_2) or (a_1, a_2, a_3) depending on whether our coordinate system is two- or three-dimensional.

→ These coordinates a_1, a_2, a_3 are called the components (scalar components) of the vector and we show

$$\vec{a} = \langle a_1, a_2 \rangle \text{ or } \vec{a} = \langle a_1, a_2, a_3 \rangle$$

Ex Find the vector represented by the directed segment with initial point $A(2, -3, 4)$ and terminal point $B(-2, 1, 1)$

$$\vec{a} = \vec{AB} = (b_1 - a_1, b_2 - a_2, b_3 - a_3)$$

$$= (-2 - 2, 1 - (-3), 1 - 4)$$

$$= (-4, 4, -3)$$

$A = (a_1, a_2, a_3)$
 $B = (b_1, b_2, b_3)$

Magnitude or length of a vector $\vec{v}$

Magnitude or length of a vector $\vec{v}$ is denoted by $|\vec{v}|$ or $\|\vec{v}\|$

The length of two dimensional vector $\vec{a} = (a_1, a_2)$ is $|\vec{a}| = \sqrt{a_1^2 + a_2^2}$

The length of three dimensional vector $\vec{a} = (a_1, a_2, a_3)$ is $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

VECTORS

→ In $\mathbb{R}^2$ we single out two particular vectors of special attention

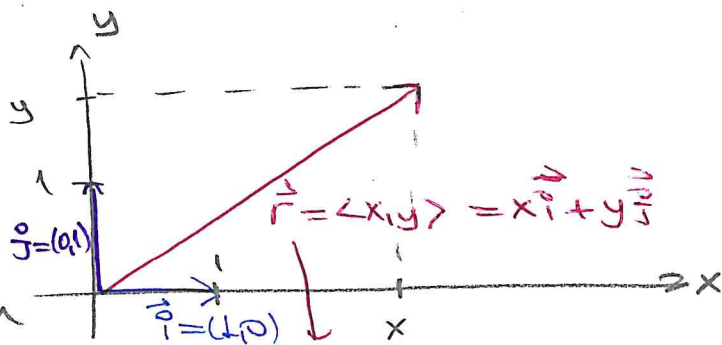
$\textcircled{1} \vec{i} = (1, 0)$
 $\textcircled{2} \vec{j} = (0, 1)$

These vectors $\vec{i}, \vec{j}$ are called standard basis vectors in the plane

→ The vector $\vec{r}$ from the origin to the point (x, y) has the components x and y can be expressed in the form

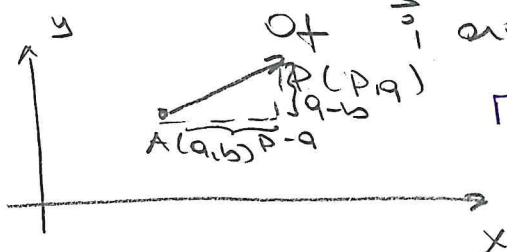
$$\vec{r} = \langle x, y \rangle = x\vec{i} + y\vec{j}$$

Linear combination



of $\vec{i}$ and $\vec{j}$.

the position vector of (x, y)



$$\vec{AP} = \langle p-a, q-b \rangle = (p-a)\vec{i} + (q-b)\vec{j}$$

ALGEBRAIC OPERATIONS ON VECTORS

→ Sums and scalar multiples of vectors are easily expressed in terms of components. Let;

$$\vec{U} = \langle u_1, u_2 \rangle = u_1 \vec{i} + u_2 \vec{j}$$

$$\vec{V} = \langle v_1, v_2 \rangle = v_1 \vec{i} + v_2 \vec{j}$$

$$\vec{U} + \vec{V} = \langle u_1 + v_1, u_2 + v_2 \rangle = (u_1 + v_1) \vec{i} + (u_2 + v_2) \vec{j}$$

$$t\vec{U} = \langle tu_1, tu_2 \rangle = tu_1 \vec{i} + tu_2 \vec{j}$$

→ The zero vector is $\vec{0} = 0\vec{i} + 0\vec{j}$. It has length zero and no specific direction. For any vector $\vec{U}$ we have $0 \in \mathbb{R}$

$$0 \cdot \vec{U} = \vec{0}.$$

UNIT VECTOR A unit vector is a vector length 1. The standard basis vectors $\vec{i}$ and $\vec{j}$ are unit vectors. Given any nonzero vector $\vec{v}$, we can form a unit vector $\hat{v}$ in the same direction as $\vec{v}$ by multiplying $\vec{v}$ by the reciprocal of its length (scalar).

$$\hat{v} = \frac{1}{|\vec{v}|} \vec{v}$$

Unit vector with the same direction $\vec{v}$

Unit vector $|\hat{v}| = 1$

Length of $\vec{v}$

Ex 7 If $A = (2, -1)$, $B = (-1, 3)$ and $C = (0, 1)$ express each of the following vectors as linear combinations of the standard basis vectors

- (a) $\vec{AB}$ (b) $\vec{BC}$ (c) $\vec{AC}$ (d) $\vec{AB} + \vec{AC}$ (e) $2\vec{AC} - 3\vec{CB}$

(f) a unit vector in the direction of $\vec{AB}$.

(a) $\vec{AB} = (-1-2)\vec{i} + (3-(-1))\vec{j} = -3\vec{i} + 4\vec{j}$

(b) $\vec{BC} = (0-(-1))\vec{i} + (1-3)\vec{j} = \vec{i} - 2\vec{j}$

(c) $\vec{AC} = (0-2)\vec{i} + (1-(-1))\vec{j} = -2\vec{i} + 2\vec{j}$

(d) $\vec{AB} + \vec{BC} = \vec{AC} = -2\vec{i} + 2\vec{j}$

(e) $2\vec{AC} - 3\vec{CB} = 2(-2\vec{i} + 2\vec{j}) - 3(-\vec{i} + 2\vec{j}) = -\vec{i} - 2\vec{j}$

(f) A unit vector in the direction $\vec{AB} \Rightarrow \vec{u} = \frac{\vec{AB}}{|\vec{AB}|} = \frac{-3\vec{i} + 4\vec{j}}{\sqrt{9+16}}$

$\vec{u} = \left(-\frac{3}{5}, \frac{4}{5}\right)$

Properties of Vector Operations

Let $\vec{u}, \vec{v}$ and $\vec{w}$ be vectors and a, b be scalars

(1) $\vec{u} + \vec{v} = \vec{v} + \vec{u}$

(2) $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$

(3) $\vec{u} + \vec{0} = \vec{u}$

(4) $(\vec{u}) + (-\vec{u}) = \vec{0}$

(5) $0 \cdot \vec{u} = \vec{0}$

(6) $1 \cdot \vec{u} = \vec{u}$

(7) $a(b\vec{u}) = (ab)\vec{u}$

(8) $a(\vec{u} + \vec{v}) = a\vec{u} + a\vec{v}$ (9) $(a+b)\vec{u} = a\vec{u} + b\vec{u}$

VECTORS IN $\mathbb{R}^3$

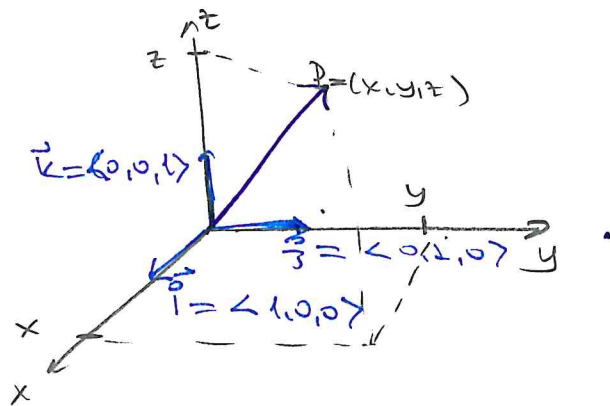
→ Given a Cartesian coordinate system in 3-space, we define three standard basis $\vec{i}, \vec{j}, \vec{k}$ where $\vec{i} = \langle 1, 0, 0 \rangle$, $\vec{j} = \langle 0, 1, 0 \rangle$, $\vec{k} = \langle 0, 0, 1 \rangle$. Any vector in $\mathbb{R}^3$ can be written as a linear combination of these basis vectors; for instance, the position vector of the point (x, y, z) is given by

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

→ we say $\vec{r}$ has components x, y and z . The length of $\vec{r}$ is

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$\vec{r} = \langle x, y, z \rangle$$



→ If $P_1 = (x_1, y_1, z_1)$ and $P_2 = (x_2, y_2, z_2)$ are two points in 3-space, then the vector $\vec{v} = \overrightarrow{P_1 P_2}$ from P_1 to P_2 has components $x_2 - x_1$, $y_2 - y_1$ and $z_2 - z_1$ and is therefore represented in terms of the standard basis vectors

by

$$\vec{v} = \overrightarrow{P_1 P_2} = (x_2 - x_1)\vec{i} + (y_2 - y_1)\vec{j} + (z_2 - z_1)\vec{k}$$

Ex 7 If $\vec{u} = 2\vec{i} + \vec{j} - 2\vec{k}$ and $\vec{v} = 3\vec{i} - 2\vec{j} - \vec{k}$ find $\vec{u} + \vec{v}$, $\vec{u} - \vec{v}$, $3\vec{u} - 2\vec{v}$, $|\vec{u}|$, $|\vec{v}|$ and a unit vector $\hat{u}$ in the direction of $\vec{u}$

$$\vec{u} + \vec{v} = (2+3)\vec{i} + (1-2)\vec{j} + (-2-1)\vec{k} = 5\vec{i} - \vec{j} - 3\vec{k}$$

$$\vec{u} - \vec{v} = (2-3)\vec{i} + (1+2)\vec{j} + (-2+1)\vec{k} = -\vec{i} + 3\vec{j} - \vec{k}$$

$$3\vec{u} - 2\vec{v} = (6-6)\vec{i} + (3+4)\vec{j} + (-6+2)\vec{k} = 7\vec{j} - 4\vec{k}$$

$$|\vec{u}| = \sqrt{4+1+4} = 3 \quad |\vec{v}| = \sqrt{9+4+1} = \sqrt{14}$$

$$\hat{u} = \left(\frac{1}{|\vec{u}|} \right) \vec{u} = \frac{\langle 2, 1, -2 \rangle}{3} = \left\langle \frac{2}{3}, \frac{1}{3}, -\frac{2}{3} \right\rangle$$

